CY4001: Industrial Chemistry

Unit 4:Polymer Chemistry
(Lecture1: History of Polymers)

CY4001: Industrial Chemistry

Suggested books

- Robert J. Young and Peter A. Lovell, Introduction to Polymers, Third Edition, CRC Press
- Fred W. Billmeyer, Text book of Polymer science, Third edition, John Wiley and sons
- The Macrogalleria - a cyberwonderland of polymer fun (https://pslc.ws/macrog/)
- (Or) Any polymer science book

Hermann Staudinger

Father of Macromolecular Chemistry.

- Hermann Staudinger was a German organic chemist who demonstrated the existence of macromolecules, which he characterized as polymers.
- He was awarded the Nobel Prize in Chemistry in 1953 for his discoveries in the field of macromolecular chemistry.

- Earlier people believed that the low molecular weight compounds "aggregated" to display characteristics like polymers rubber.

- Staudinger showed that such compounds were actually covalently bonded molecules of high molecular weight formed by the linking together of smaller molecules. *But the world of science took a long time to agree with him.*

Staudinger openly criticized his native country's involvement in the war

Unit 4:Polymer Chemistry

What are Polymers and why are they interesting?

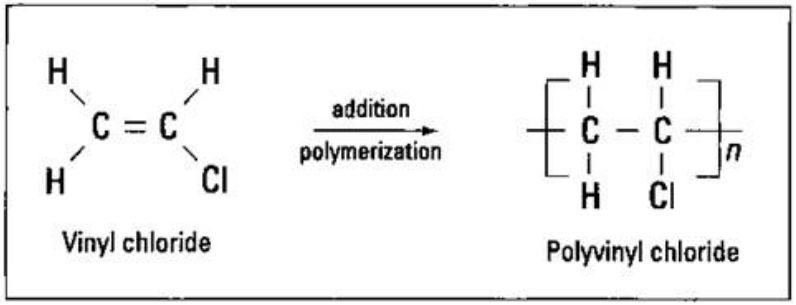
Polymeric Products

Polymer clay train

Polymer toy train

Polymer train floor

All the above products are based on same type of polymer called polyvinyl chloride (PVC)



Unit 4:Polymer Chemistry

Nobel Prizes awarded for research in the field of polymer science

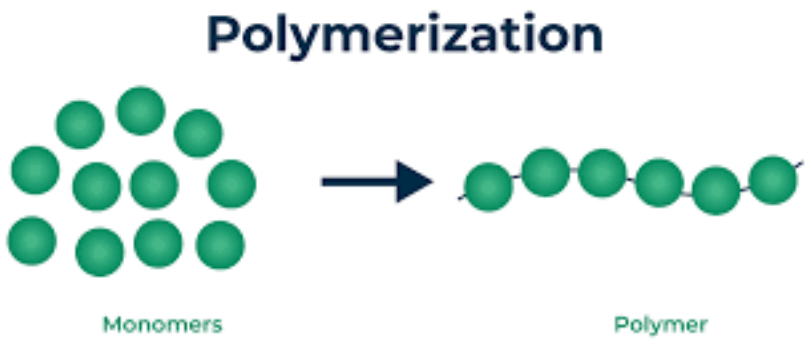
Researchers	Year	Area
Hermann Staudinger	1953	“for his discoveries in the field of macromolecular chemistry”
Karl Ziegler, Giulio Natta	1963	"for their discoveries in the field of the chemistry and technology of high polymers"
Paul J. Flory	1974	“for his fundamental achievements, both theoretical and experimental, in the physical chemistry of the macromolecules”
Pierre-Gilles de Gennes	1991	"for discovering that methods developed for studying order phenomena in liquid crystals and polymers"
Alan Heeger, MacDiarmid, Shirakawa	2000	"for the discovery and development of conductive polymers"
John Fenn, Koichi Tanaka & Wüthrich	2002	"for the development of methods of biological macromolecules"
Y Chauvin, Robert Grubbs & Schrock	2005	"for the development of the metathesis method in organic synthesis"

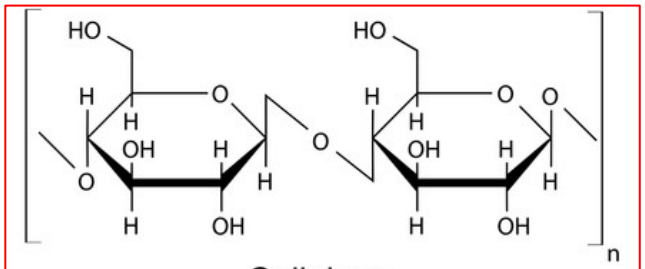
Unit 4:Polymer Chemistry

Terminology

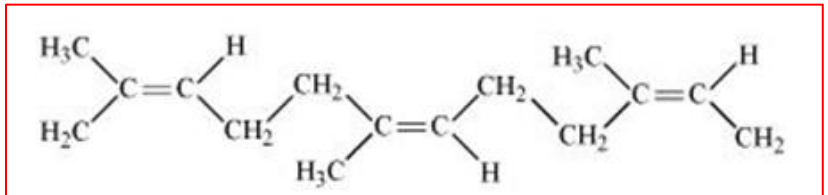
- **The word polymer is derived from the Greek words *poly* meaning 'many' and *mers* meaning 'units' or 'parts'.**

- Polymers are generally macromolecules formed by the repeated linking of a large number of small molecules. The chemistry of these giant molecules is known as 'polymer chemistry' or 'macromolecular chemistry'.



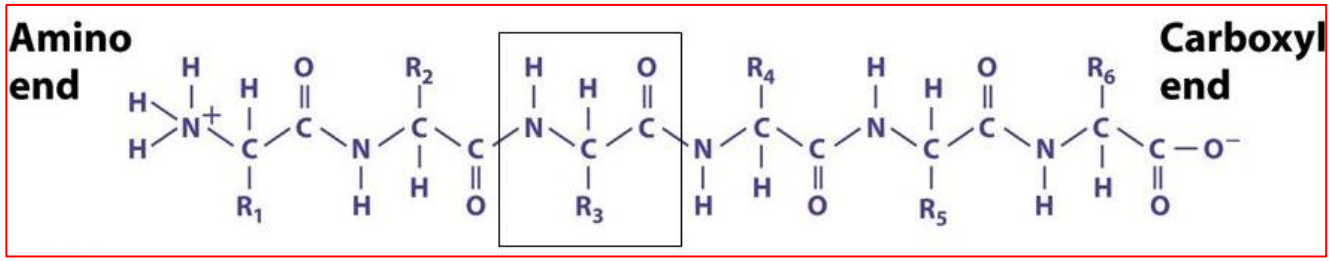


Cellulose



Natural rubber

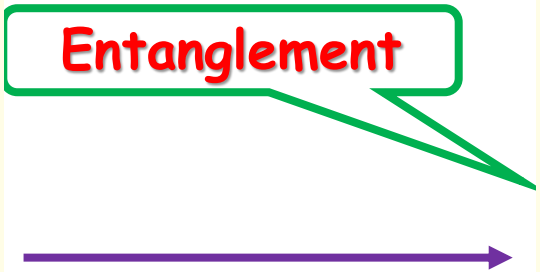
Number of repeating units – 100 to 100000



Unit 4:Polymer Chemistry

Properties of Polymers: Consequences of long chain formation

Monomer (rice)



Polymer (rice noodles)

High molecular weight polymers have long chains that make it difficult for them to pass each other. When they try to squeeze past each other, they become entangled. This results in a higher viscosity and physical properties.

Semi-crystalline

Amorphous

Unit 4:Polymer Chemistry

Properties of Polymers: Consequences of long chain formation

(1) Entanglement: It strongly reduced the chain mobilities.

(2) Glass transition temperature: Polymer exhibits Glass transition temperature. Mobility of chains depends on temperature.

Glass transition temperature (Tg) is the temperature at which an amorphous solid material transforms from a hard, brittle state to a softer, more malleable state. This phenomenon occurs because the bonds between molecular chains become less rigid and more mobile

(3) Viscoelastic properties: Polymer exhibits combined properties of liquids (viscosity) and solids (elasticity)

(4) Semicrystalline properties: Polymer exhibits combined properties of crystalline and amorphous materials

Thus polymers are interesting!

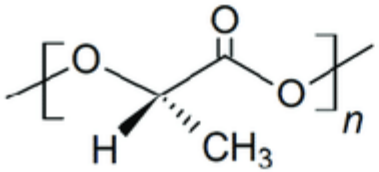
Unit 4:Polymer Chemistry

Properties of Polymers: General

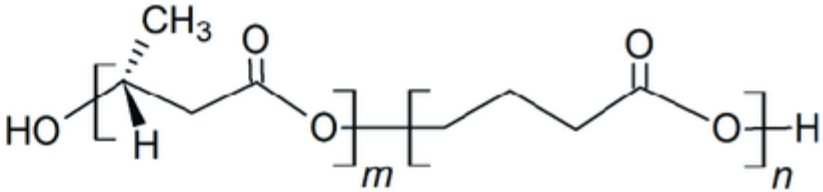
- Easy synthetic methods (low temperature)
- Low cost manufacturing (from renewable sources, petrochemicals.. etc)
- Light weight materials (Glass bottle – 100 g; plastic bottle – 10 g)
- Readily processable (molding, printing etc.)

Biodegradable??

BIODEGRADABLE POLYMERS



Polylactide (PLA)



Polyhydroxyalkanoate (PHA)

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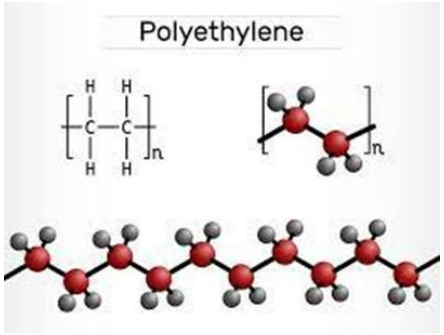
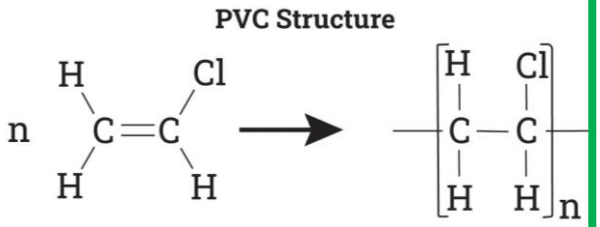
Some applications

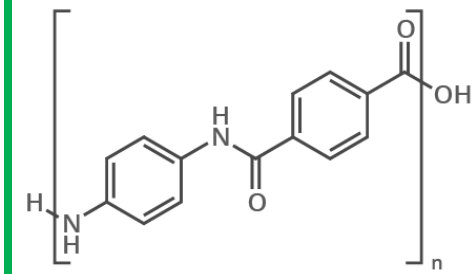
low cost, and good moisturebarrier (Packaging)



High-strength thermoplastic (PVC pipes)



Strong, lightweight, and resistant to impact (bullet proof jacket)



Poly-p-phenylene terephthalamide (PPTA)

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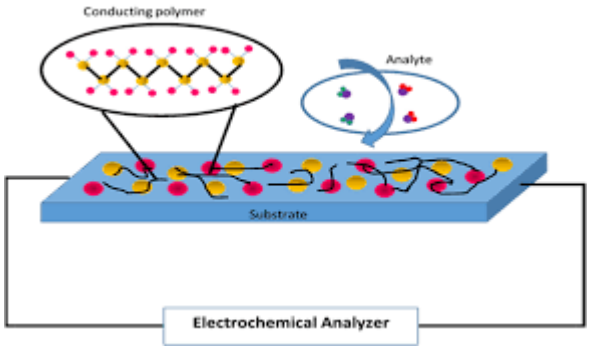
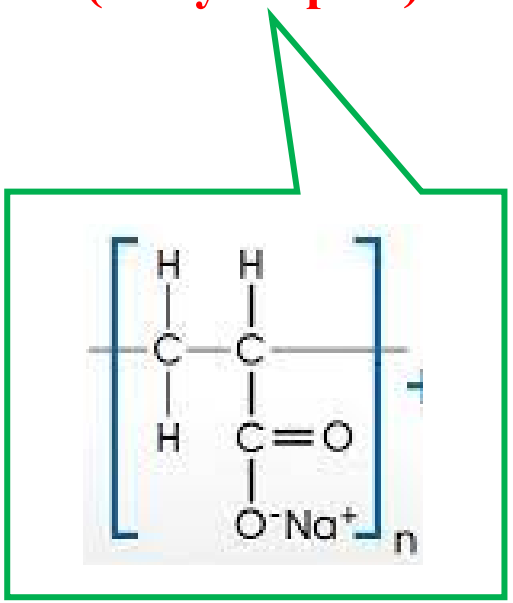
Some applications

Superabsorbents (Baby diapers)

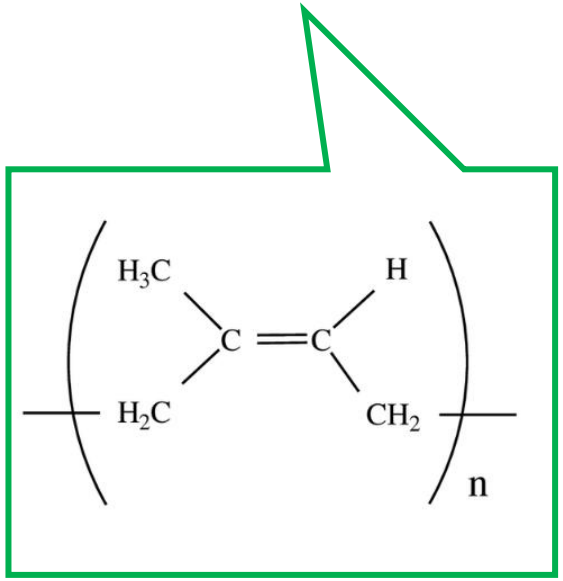
Insulators (polymer electrical insulators)



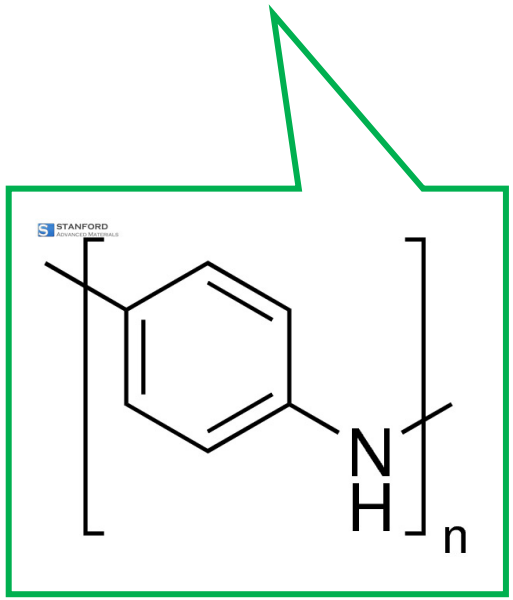
Conductors (sensors)



Sodium polyacrylate



Polyisoprene



Polyaniline

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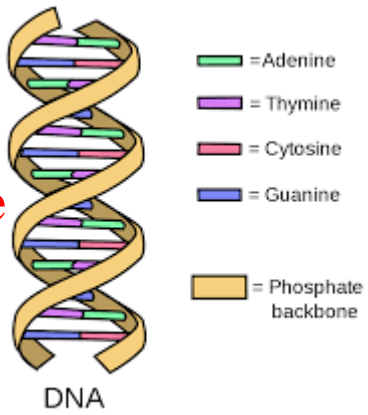
History of Polymers

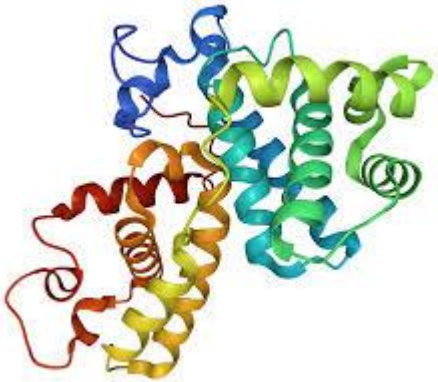
1000000 BC.. No life without polymers!! why?

DNA is a poly-nucleobase



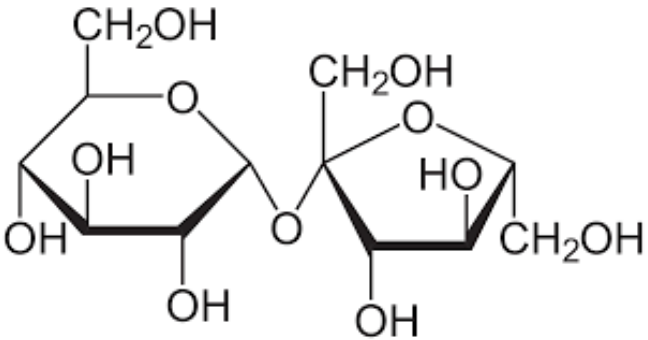
Information storage and transfer

Proteins are poly-aminoacids



Proteins are functional machines of our body

Carbohydrates are poly-saccharides



Sources of energy

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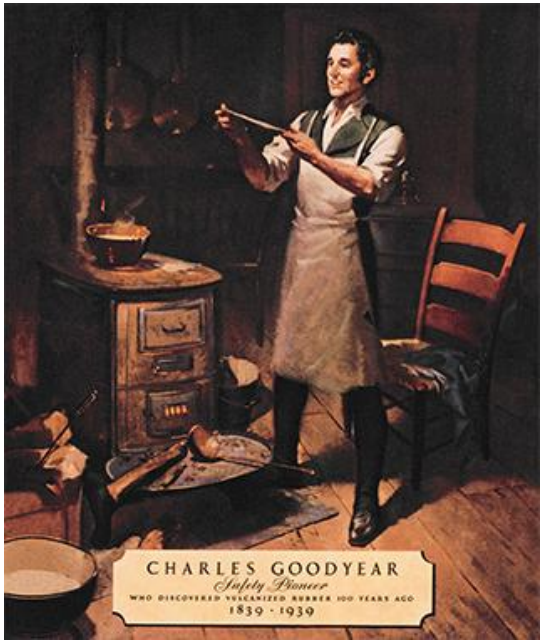
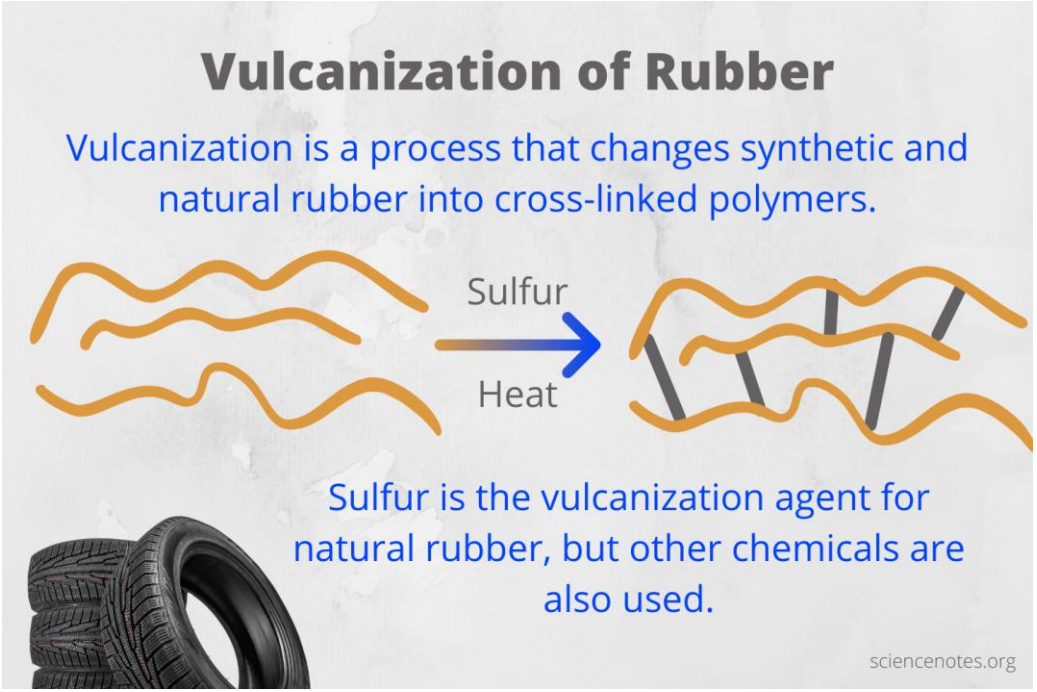
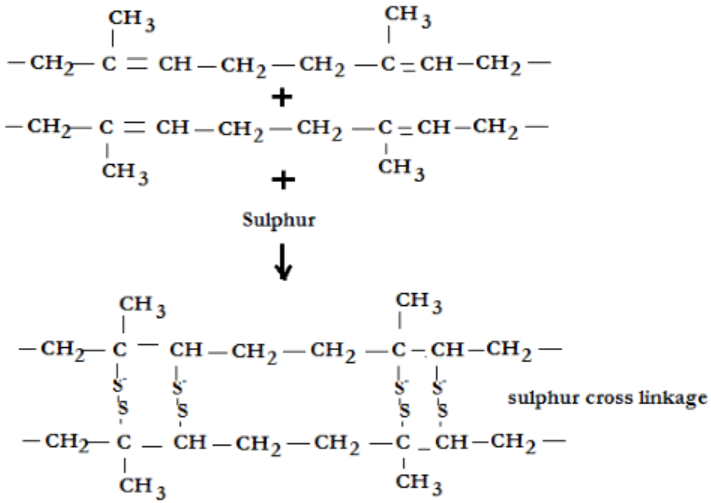
History of Polymers

1839: Vulcanization – Charles Good year

Charles Goodyear saved the doomed rubber industry by inventing a process that made the material durable and resilient enough for industrial use.







Kickstarter for the automotive industry

Unit 4:Polymer Chemistry

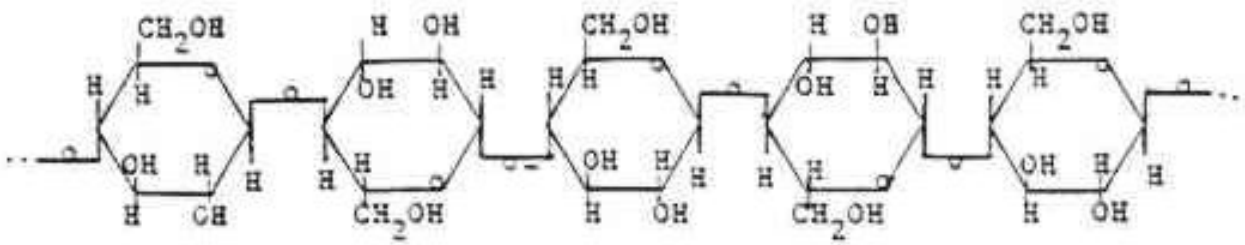
History of Polymers

1860: Man-made Polymers (Invention of Celluloid)

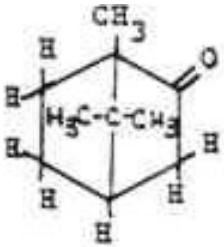
- During the Victorian era, it was fashionable for wealthy gentlemen to own a billiard table and a set of billiard balls crafted of the finest and most perfect ivory. By 1863, the ivory shortage had become so critical that a New York billiard-ball manufacturer offered a $10,000 prize to the person who could create a useful substitute.

John Wesley Hyatt invented celluloid as an alternative to ivory and saved thousands of elephants. Celluloid is a hard solid at normal temperatures and when heated it became soft and could be molded in to desired shapes

Hyatts made celluloid by applying heat and pressure to a mix of cellulose nitrate and camphor; it was thus a plastic made by modifying natural material



CELLULOSE



CAMPHOR

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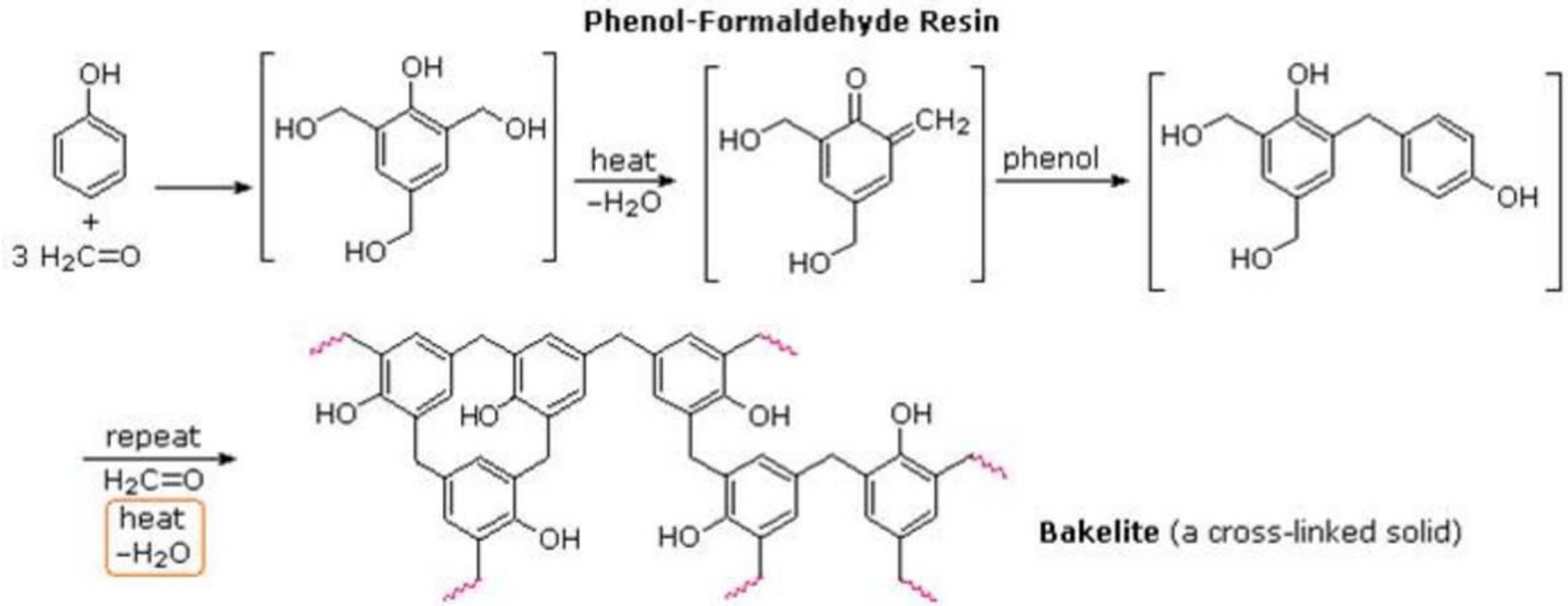
History of Polymers

1910: Bakelite, the world's first completely synthetic plastic

1910 by Leo Baekeland, invented Bakelite, a hard polymer resin from phenol and formaldehyde



Unit 4:Polymer Chemistry

Bakelite (1910): the world's first completely synthetic plastic

Bakelite Properties:

- It is a hard, rigid, infusible, water resistant and insoluble solid.
- Bakelite is resistant to acids, salts and most organic solvents, but it is attacked by alkalies because of the presence of –OH groups.
- It acts as an excellent electrical insulators.
- It can withstand high temperature.

Bakelite Uses:

- It is used for making moulded articles such as radio and TV parts, etc.
- It is used as an adhesive in plywood laminations and in grinding wheels, etc.
- It is commonly used in paints and varnishes.
- It is used for making electrical insulation parts like plugs, switches, heater handles, etc.

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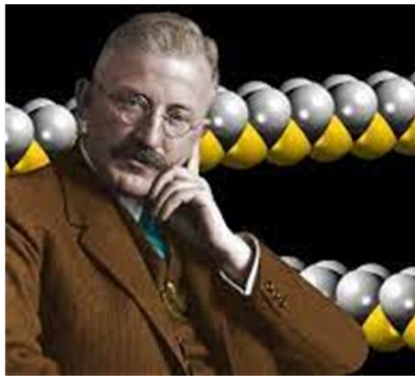
History of Polymers



1924: Macromolecular hypothesis

Hermann Staudinger proposed the macromolecular hypothesis in 1924, and won Nobel prize in 1953.

Scientific Problem of polymeric substances (at that time)

- Cannot be distilled
- Viscous in nature
- Do not have any end groups
- Do not properly crystalize

Hermann Staudinger proposed the macromolecular hypothesis, which states that polymers are molecules made of covalently bonded elementary units called monomers.

Outcome from the hypothesis,

- Concept of macromolecules
- Determination of molecular weights

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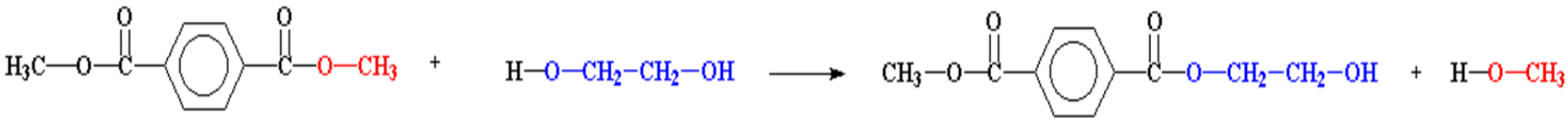
History of Polymers

1928: First rational synthesis of macromolecules

1928, Wallace H. Carothers, built up polymers by reacting small organic molecules by means of well-known reactions to form long, macromolecular chains



1939-1945: World war (II) boosted polymer production

Nylon – parachutes
Buna (Synthetic rubber) - tyres